MD SHAHFAHAD KHAN

Machine Learning Engineer | AI/ML

+91-8928530995 | khannishu522@gmail.com | LinkedIn | GitHub | Portfolio | Delhi

SUMMARY

BCA graduate and Machine Learning Engineer with hands-on experience building end-to-end AI/ML and Generative AI applications using Python, PyTorch, Scikit-learn, FastAPI, Flask, and Docker. Skilled in Machine Learning, Deep Learning, NLP, Computer Vision, Retrieval-Augmented Generation (RAG), and Semantic Search, with a track record of deploying production-ready ML and LLM-powered solutions.

ACADEMIC PROJECTS

AI Research Assistant (RAG-based) | *Jul 2026* | GitHub | Live Demo

Tech Stack: *Python • FastAPI • Gemini API • ChromaDB • RAG • PyMuPDF • HTML/CSS • Railway*

- Built an end-to-end AI Research Assistant using Retrieval-Augmented Generation (RAG) to enable intelligent, citation-backed question answering over uploaded PDF documents.
- Designed the RAG pipeline covering PDF parsing, text chunking, semantic embeddings, and vector similarity search using ChromaDB combined with the Gemini API for grounded answer generation.
- Developed and deployed a FastAPI application with document upload, knowledge base management, and an end-to-end AI-powered research workflow, hosted on Railway.

DeepRank AI – Resume–Job Semantic Ranking Engine | *Jul 2026* | GitHub | Live Demo

Tech Stack: *Python • Flask • PyTorch • Sentence-BERT (SBERT) • NLP • Semantic Search • Scikit-learn • Docker*

- Engineered a deep learning-based semantic resume ranking engine using Sentence-BERT (SBERT) to intelligently rank 500+ resumes against job descriptions.
- Built an AI recruitment platform featuring semantic search, weighted skill-gap analysis, duplicate resume detection, candidate recommendation, and explainable AI insights.
- Developed scalable resume processing, a recruiter analytics dashboard, and downloadable CSV/PDF/ZIP reports; deployed on Hugging Face Spaces using Docker.
- Optimized transformer-based semantic retrieval for accurate candidate ranking beyond traditional keyword matching.

Customer Churn Prediction & Retention Recommendation System | *Jul 2026* | GitHub | Live Demo

Tech Stack: *Python • Scikit-learn • Flask • Pandas • NumPy • Matplotlib*

- Built an end-to-end customer churn prediction system using demographic, service usage, contract, and billing data.
- Performed EDA, preprocessing, feature engineering, encoding, and scaling; tuned and evaluated Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Developed and deployed a Flask application that predicts churn probability, classifies customers into Low, Medium, and High Risk tiers, and generates personalized retention recommendations.

House Rent Price Prediction System | *Jan 2026 – Mar 2026* | GitHub | Live Demo

Tech Stack: *Python • Scikit-learn • Random Forest • Flask • Pandas • NumPy • SQLite*

- Developed an end-to-end Random Forest-based web application to predict house rental prices from real-world housing datasets.
- Built a complete preprocessing pipeline including missing value imputation, feature engineering, categorical encoding, outlier removal, and hyperparameter tuning.
- Achieved approximately 0.75 R^2 score and deployed a production-ready Flask application with secure user authentication and an admin dashboard.

EDUCATION

Chandigarh University

Bachelor of Computer Applications (BCA) | Graduated June 2026 | CGPA: 7.50/10

TECHNICAL SKILLS

Programming: Python • SQL • C++

Machine Learning: Regression • Classification • Ensemble Learning • Feature Engineering • Hyperparameter Tuning • Model Evaluation

Deep Learning & GenAI: PyTorch • TensorFlow • CNN • Transfer Learning • Computer Vision • NLP • RAG • Semantic Search • LLM APIs (Gemini)

Frameworks & Tools: Flask • FastAPI • REST APIs • Docker • ChromaDB • Hugging Face Spaces • Railway • Render

Libraries: Scikit-learn • NumPy • Pandas • OpenCV • PyMuPDF • Matplotlib • Seaborn

Databases: MySQL • PostgreSQL • ChromaDB (Vector DB)

Dev Tools: Git • GitHub • Linux • VS Code • Jupyter Notebook